

CHAPTER 9

Excitons, Biexcitons and Trions

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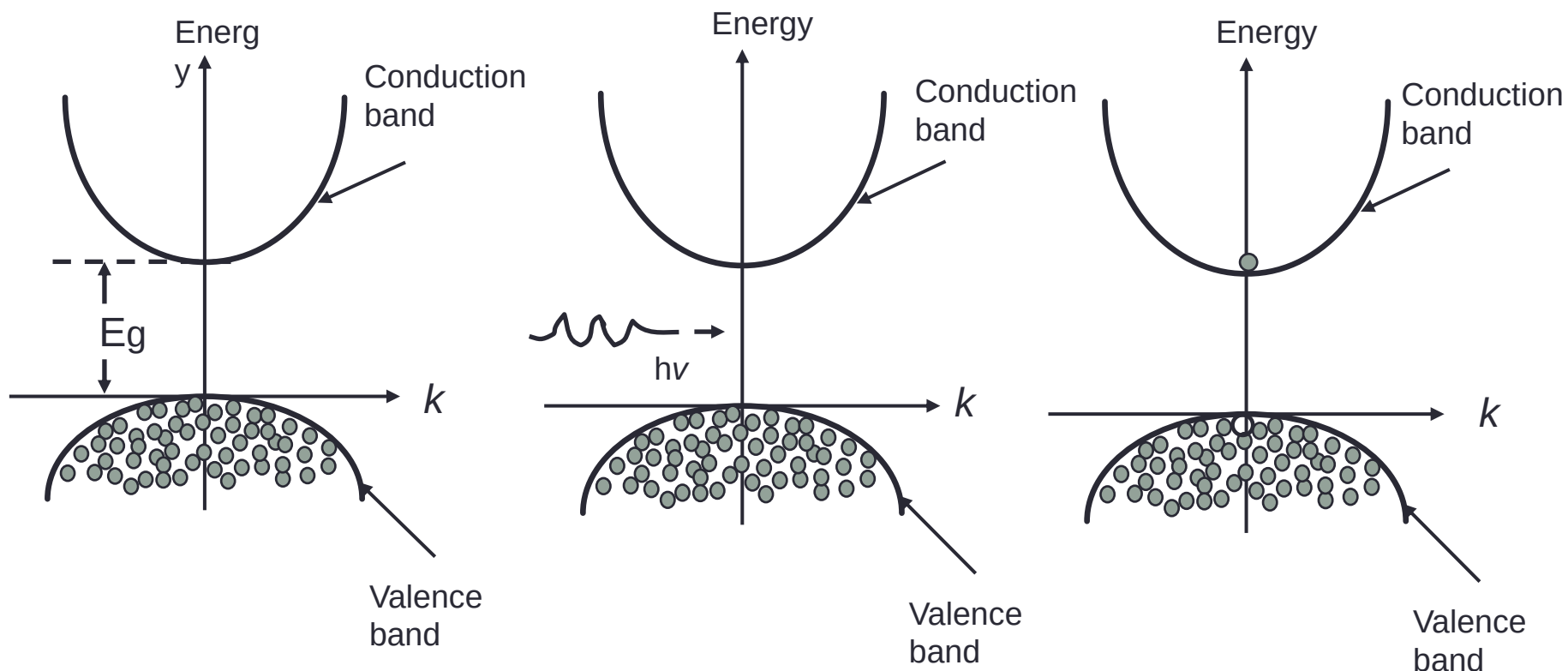
- In this chapter we introduce the concepts of Exciton, Biexciton and Trion.
- We discuss their behaviors in bulk semiconductors and semiconductors with reduced dimensionality.

Exciton

Ground state: Valence band is completely full, conduction is empty

Excited state: Excite electrons from valence band to conduction band by absorption of photons

Exciton: A bound state of an electron and a hole formed in an insulator or semiconductor through Coulomb interaction



9.1 Wannier and Frenkel Excitons

- Fig.9.1a: Coulomb interaction: with a Coulomb potential term $-e^2/(4\pi\epsilon_0\epsilon|\mathbf{r}_e - \mathbf{r}_h|)$
- Fig.9.1b: The dispersion relation of excitons :

$$E_{ex}(n_B, \mathbf{K}) = E_g - Ry^* \frac{1}{n_B^2} + \frac{\hbar^2 \mathbf{K}^2}{2M}$$

with

$n_B = 1, 2, 3, \dots$ principal quantum number

$$Ry^* = 13.6 \text{ eV} \frac{\mu}{m_0} \frac{1}{\epsilon^2} \quad \text{exciton Rydberg energy}$$

$M = m_e + m_h$, $\mathbf{K} = \mathbf{k}_e + \mathbf{k}_h$ translational mass and wave vector of the exciton

$$\mu = \frac{m_e m_h}{m_e + m_h} \quad \text{reduced exciton mass}$$

$$a_B^{\text{ex}} = a_B^{\text{H}} \frac{m_0}{\mu} \quad \text{excitonic Bohr radius.}$$

Using the material parameters for typical semiconductors :

$$1 \text{ meV} \leq Ry^* \leq 200 \text{ meV} \ll E_g$$

$$50 \text{ nm} \geq \alpha_B \geq 1 \text{ nm} > \alpha_{\text{lattice}}$$

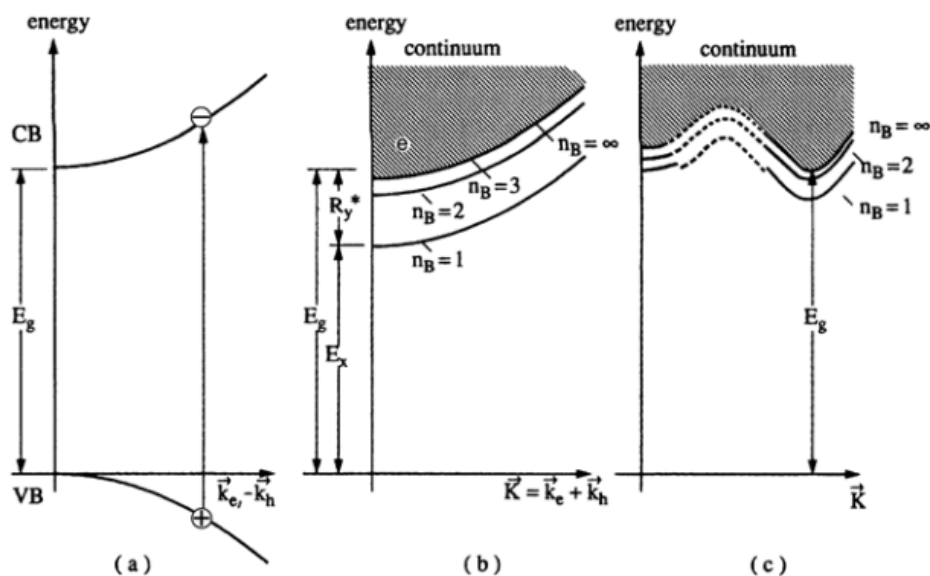
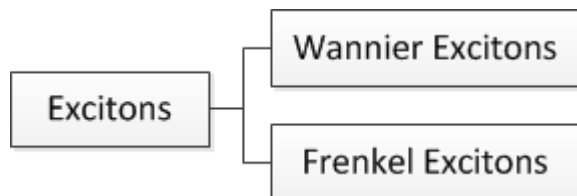


Fig. 9.1. A pair excitation in the scheme of valence and conduction band (a) in the exciton picture for a direct (b) and for an indirect gap semiconductor (c)

- For Wannier excitons , the exciton wavefunction:

$$\phi(\mathbf{K}, n_B, l, m) = \Omega^{-1/2} e^{i\mathbf{K} \cdot \mathbf{R}} \phi_e(\mathbf{r}_e) \phi_h(\mathbf{r}_h) \phi_{n_B, l, m}^{env}(\mathbf{r}_e - \mathbf{r}_h) \quad (9.4a)$$

with the center of mass $\mathbf{R}$: $\mathbf{R} = (m_e \mathbf{r}_e + m_h \mathbf{r}_h) / (m_e + m_h)$

For direct semiconductors : $f_{nB} \propto |H_{cv}^D|^2 \frac{1}{n_B^3}$ (9.5)

- The electron–hole pair, likewise the exciton, is said to be in a spin singlet state.
- Spin triplet state: $S=1$ (if the spin flips in the transition), situated energetically below the singlet state.
- In the picture of second quantization, $\alpha_{k_e}^+$, $\beta_{k_h}^+$ for the creation operators of electrons in the conduction band and holes in the valence band.

so creation operators for electron hole pairs : $\alpha_{k_e}^+ \beta_{k_h}^+$

The exciton creation operator : $B_K^+ = \sum_{k_e, k_h} \delta[\mathbf{K} - (\mathbf{k}_e + \mathbf{k}_h)] \alpha_{k_e}^+ \beta_{k_h}^+$ (9.6)

- Obey Bose commutation relations.
- In thermodynamic equilibrium for low densities: the excitons can be well described by Boltzmann statistics .
- For higher densities: they deviate more and more from ideal bosons until they end up in an electron–hole plasma made up entirely from fermions.

9.2 Corrections to the Simple Exciton Model

- A first group of corrections relevant for ε and μ .
- For GaAs : $E_{ex}^b < \hbar\omega_{LO}$; $a_B > a_{Pol} \Rightarrow \varepsilon = \varepsilon_s$
- For CdS, ZnO.....: $E_{ex}^b \geq \hbar\omega_{LO}$; $a_B ; a_{Pol} \rightarrow \varepsilon_s \geq \varepsilon \geq \varepsilon_b$

So using Haken potential :

$$\frac{1}{\varepsilon(\mathbf{r}_{e,h})} = \frac{1}{\varepsilon_b} - \left(\frac{1}{\varepsilon_b} - \frac{1}{\varepsilon_s} \right) \left(1 - \frac{\exp(-\mathbf{r}_{eh}/r_e^P) + \exp(-\mathbf{r}_{eh}/r_h^P)}{2} \right) \quad (9.8)$$

◆ An increase of the exciton binding energy with increasing E_g

Fig. 9.2 The splitting of the 1S exciton with the notation of the various contributions to exchange splitting for bulk samples (a) and in a quantum dot (b). The abbreviations have the following meanings: S = singlet, T = triplet, Δ^{lr} = long range or nanoanalytic (for $k \rightarrow 0$) part of the exchange interaction, Δ^{sr} = short range or analytic part of the exchange interaction [03G1]

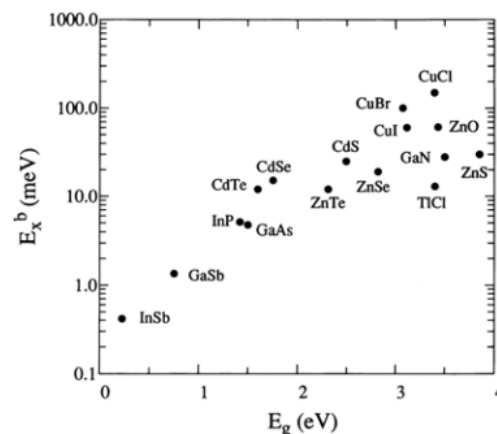
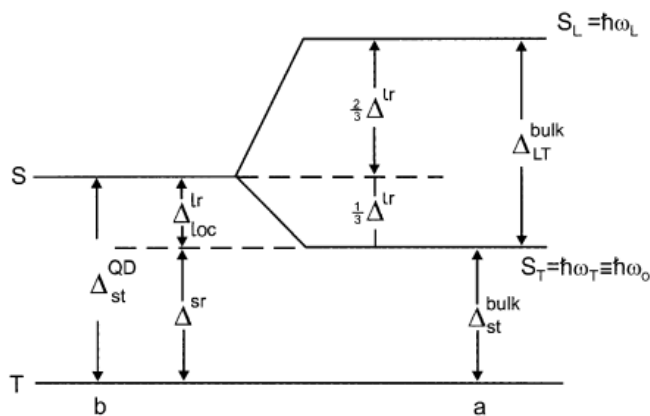


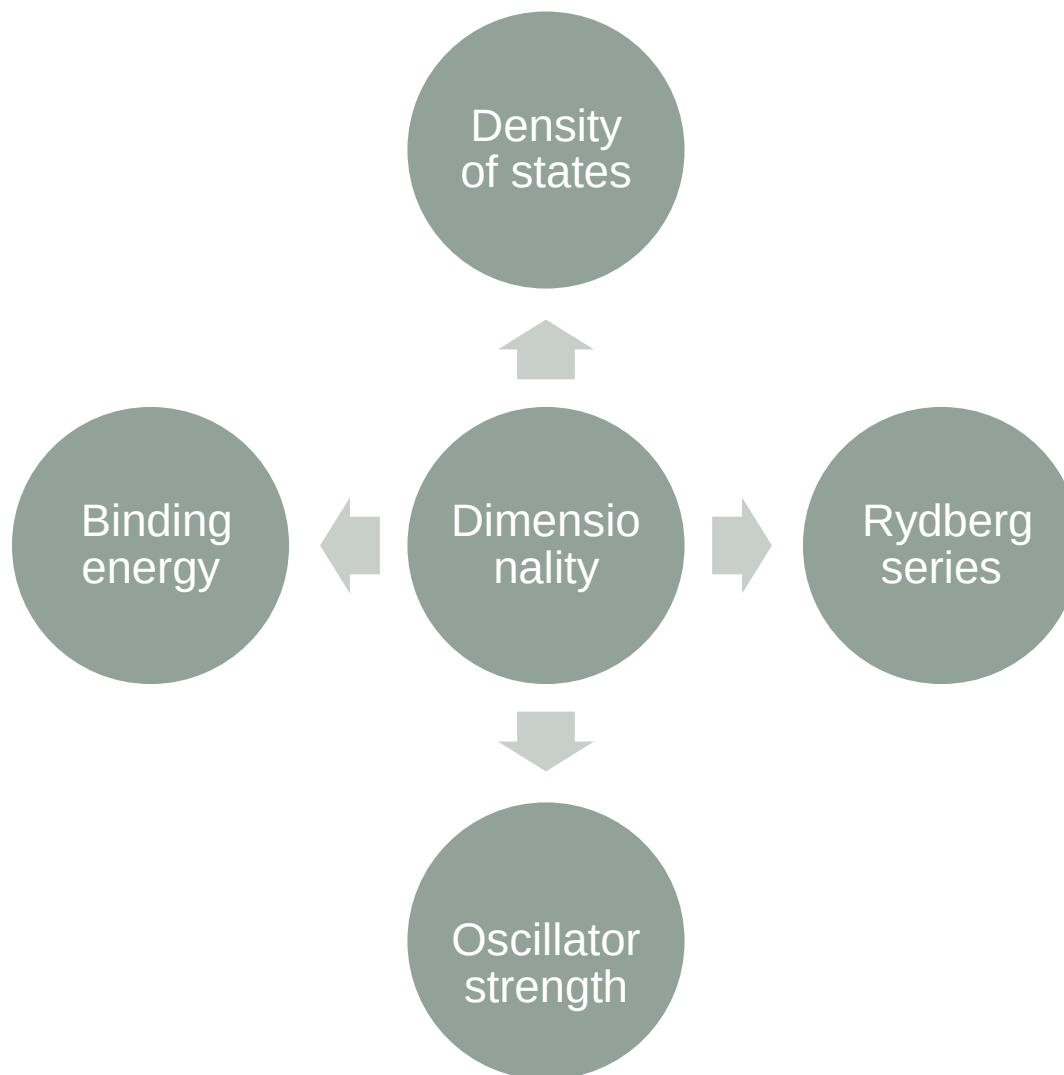
Fig. 9.3. The exciton binding energy E_{ex}^b as a function of the band-gap for various direct gap semiconductors ([82L1, 93H1, 93P1] of Chap. 1)

- Exchange interaction between electron and hole caused by their Coulomb interaction result in the splitting between singlet and triplet excitons Δ_{st} and the splitting of singlet state into a transverse and a longitudinal one Δ_{LT}
- The following relation holds for Wannier excitons:

$$\Delta_{st} = \Delta_{LT} \quad \text{with} \quad 0.1\text{meV} \leq \Delta_{LT} \leq 15\text{meV} \quad (9.9)$$

- The singlet-triplet splitting is even enhanced by the compression of the excitonic wave function in quantum wells and especially in quantum dots.
- The excitation of an optically allowed exciton is accompanied by a polarization.
- The excitons can also be formed with holes in deeper valence bands.

9.3 The Influence of Dimensionality



- We consider an exciton:

3D :

$$3d: E(\mathbf{K}, n_B) = E_g - Ry^* \frac{1}{n_B^2} + \frac{\hbar^2(K_x^2 + K_y^2 + K_z^2)}{2M} \quad (9.10a)$$

For the oscillator strength f :

$$f(n_B) \propto n_B^{-3}; \quad a_B \propto a_B^H n_B; \quad n_B = 1, 2, 3K \quad (9.10b)$$

2D :

$$2d: E(\mathbf{K}, n_B) = E_g + E_Q - Ry^* \frac{1}{\left(n_B - \frac{1}{2}\right)^2} + \frac{\hbar^2(K_x^2 + K_y^2)}{2M} \quad (9.10c)$$

with E_Q quantization energy:

$$f(n_B) \propto \left(n_B - \frac{1}{2}\right)^{-3}; \quad a_B \propto a_B^H \left(n_B - \frac{1}{2}\right); \quad n_B = 1, 2, 3K \quad (9.10d)$$

	3D $\rightarrow$ 2D
n_B	$n_B \rightarrow (n_B - 1/2)$
Ry^*	The same
Oscillator strength	Increases
Excitonic Bohr radius	Decreases

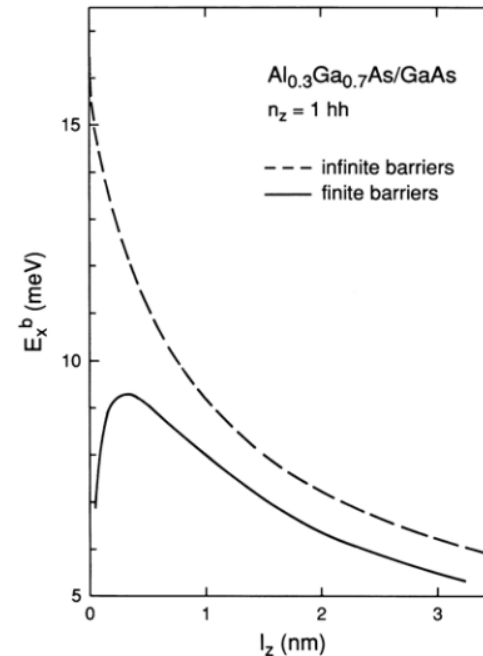


Fig. 9.4. The calculated binding energy of $n_z = 1$ hh excitons in AlGaAs/GaAs quantum wells as a function of the well thickness l_z ([81M1, 84G1, 88K2])

- The binding energy of the exciton in the quasi two-dimensional case :

$$E_{ex}^b(1S) = \frac{Ry^*}{\left(1 + \frac{d_{eff} - 3}{2}\right)^2} \quad \text{with} \quad d_{eff} = 3 - \exp\left(\frac{-Lw}{2a_\beta}\right) \quad (9.10)$$

The Sommerfeld factor F: describes the enhancement of the oscillator strength of the continuum states .

$$3d : F_{3d} = \frac{\pi}{W^{1/2}} \frac{e^{\pi W^{1/2}}}{\sinh(\pi W^{1/2})} \quad \text{with} \quad W = (E - E_g) Ry^{*-1} \quad (9.11)$$

$$2d : F_{2d} = \frac{e^{\pi W^{1/2}}}{\cosh(\pi W^{1/2})} \quad \text{with} \quad W = (E - E_g + E_Q) Ry^{*-1}$$



- For quantum dots:

Weak confinement: $R \geq a_B, E_Q < Ry^*$

Medium or intermediate confinement : $a_B^e \geq R \geq a_B^h; E_Q \approx Ry^*, \text{ with } a_B^{e,h} = a_B^H \frac{m_0}{m_{e,h}}$

Strong confinement: $R \leq a_B^h, E_Q > Ry^*$

9.4 Biexcitons and Trions

- Biexciton or excitonic molecule: a new quasiparticle formed by two excitons. The dispersion relation is given :

$$E_{biex}(\mathbf{k}) = 2(E_g - E_{ex}^b) - E_{biex}^b + \frac{\hbar^2 \mathbf{k}^2}{4M_{ex}} \quad (9.14)$$

Biexcitons also form bound states in quantum wells, wires and dots again in agreement with experiments.

- Trions: charged excitons or biexcitons, i.e., quasiparticles consisting of two electrons and one hole or vice versa.

For quantum structures: trions also form bound states contributing to the luminescence below the free exciton energy.

9.5 Bound Exciton Complexes

Excitons can also be bound to defects.

- Shallow impurities.

- The binding energy of an exciton is highest for a neutral acceptor, lower for a neutral donor and lower still for an ionized donor .
- An ionized acceptor does not usually bind an exciton .
- The absorption and emission lines of A^0X , D^0X and D^+X are often labelled I_1 , I_2 and I_3 lines, respectively.
- The binding energy of an exciton to a neutral donor (acceptor) is usually much smaller than the binding energy of an electron (hole) to the donor (acceptor).

- Donor–acceptor pairs can be considered as “polycentric” bound excitons.

- One center can, under certain conditions, bind several excitons.

- In indirect semiconductors
- In quantum wells

9.6 Excitons in Disordered Systems

We consider the two-particle complex exciton localized in a disordered semiconductor.

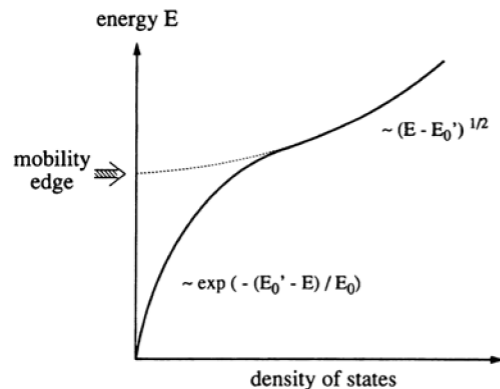


Fig. 9.5. The density of exciton states for $n_B = 1$ in a disordered semiconductor

- ▣ Low energies: start with excitons for which one carrier is localized and the other bound to it by Coulomb attraction.
- ▣ With increasing energy: a continuous transition to excitons localized as a whole. The mobility “edge”.

■ Features:

- Localized excitons more frequently in those ionic semiconductors with alloying in the anions that form the valence band like $\text{CdS}_{1-x}\text{Se}_x$ or $\text{ZnSe}_{1-x}\text{Te}_x$ than in cation-substituted materials such as $\text{Zn}_{1-y}\text{Cd}_y\text{Se}$.
- The exciton averages over a larger volume, thus diminishing the effective fluctuation and reducing the tail of localized exciton states.
- In quantum wells various types of disorder can contribute to the formation of tails of localized exciton states as already discussed for one-particle states:
 - Alloy disorder if the well and/or the barrier material is an alloy.
 - Interface roughness, i.e., well-width fluctuations.